

figureidea

Paper Outline: Rayfield

Physics-Embedded Flow Matching for Photonic Device Field Prediction

Paper Structure Guide

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1 1. Title & Abstract

1.1 Title Options

- “Rayfield: Physics-Embedded Flow Matching for Electromagnetic Field Prediction in Photonic Integrated Circuits”
- “Complex-Valued Flow Matching with Helmholtz Constraints for Photonic Device Simulation”
- Shorter: “Physics-Constrained Generative Modeling for Photonic Field Prediction”

1.2 Abstract (~150–250 words)

1. **Problem:** FDTD simulation of photonic devices is slow (minutes–hours per geometry), limiting design-space exploration and inverse design.
2. **Gap:** Existing surrogate models either ignore physics (pure data-driven) or are limited to simple geometries. No prior work combines flow matching + complex-valued networks + Helmholtz constraints for photonic simulation.
3. **Method:** We present a complex-valued Physics UNet trained via conditional flow matching with a multi-loss curriculum (FM $\rightarrow$ Helmholtz residual $\rightarrow$ S-parameters). Native complex arithmetic preserves phase. Helmholtz residual loss enforces Maxwell’s equations.
4. **Results:** Quote key metrics — PSNR, SSIM, S-parameter accuracy, and speedup vs. FDTD. Mention number of device types and dataset size.
5. **Impact:** Enables real-time field prediction for inverse design of photonic circuits.

2 2. Introduction

2.1 2.1 Motivation & Context (~1 page)

- Photonic integrated circuits (PICs) are critical for telecom, sensing, quantum computing, AI accelerators.
- Design relies on electromagnetic simulation (FDTD, FEM) — accurate but computationally expensive.
- Growing need for fast surrogate models to enable: (a) large-scale design-space exploration, (b) gradient-based inverse design, (c) real-time design feedback.

2.2 2.2 Limitations of Prior Work

- **Physics-informed neural networks (PINNs):** Require per-geometry optimization; don’t generalize across geometries.
- **Data-driven surrogates:** CNNs/U-Nets that predict fields directly — ignore physics, poor generalization, phase errors.
- **Neural operators (FNO, DeepONet):** Promising but typically real-valued; lose phase information critical for coherent photonics.

- **Generative models:** Diffusion/flow matching used in other domains but not yet applied to EM field prediction with physics constraints.

2.3 2.3 Contributions

1. **Complex-valued flow matching:** First application of conditional flow matching with native complex arithmetic to EM field prediction.
2. **Helmholtz-constrained training:** Physics residual loss that enforces $\nabla^2 E + k_0^2 \varepsilon E = 0$ with interface-aware masking.
3. **Multi-loss curriculum:** Phased training (FM $\rightarrow$ physics $\rightarrow$ S-params) that stabilizes convergence.
4. **Multi-device generalization:** Single model trained across 5+ photonic device types with varying port counts and physics.
5. **Dataset:** Large-scale multi-device FDTD dataset (10,000 geometries $\times$ 4 wavelengths = 40,000 simulations) with standardized format.

3 3. Background

3.1 3.1 Electromagnetic Simulation of Photonic Devices

- Maxwell’s equations in 2D TE polarization $\rightarrow$ scalar Helmholtz: $\nabla^2 E_z + k_0^2 \varepsilon(x, y) E_z = 0$.
- FDTD method: time-stepping on Yee grid, converges to frequency-domain solution.
- Key outputs: complex field $E_z(x, y)$, permittivity map $\varepsilon(x, y)$, S-parameters.
- Silicon photonics material system: Si core ($n_{\text{eff}} \approx 2.4$), SiO₂ cladding ($n = 1.444$), C-band wavelengths (1500–1600 nm).

3.2 3.2 Conditional Flow Matching

- Flow matching as an alternative to diffusion: learns a velocity field $u_t(x_t, t)$ that transports noise $x_0 \sim \mathcal{N}(0, I)$ to data x_1 .
- Interpolation path: $x_t = (1 - (1 - \sigma)t) x_0 + t x_1$.
- Training objective: $\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, x_0, x_1} [\|u_\theta(x_t, t) - u_t\|^2]$.
- Advantages: simpler training than diffusion, deterministic sampling via ODE integration.
- Cite: Lipman et al. (2023), Liu et al. (2023).

3.3 3.3 Physics-Informed Neural Networks

- Brief survey of PINNs, physics-constrained learning.
- Distinction: PINNs optimize per-instance; our approach trains a generalizable model with physics loss as regularization.
- Cite: Raissi et al. (2019), and relevant photonics PINN papers.

4 4. Method

4.1 4.1 Problem Formulation

- **Input:** Permittivity map $\varepsilon(x, y)$ [1,H,W], source mask [1,H,W], wavelength λ .
- **Output:** Complex field $E_z(x, y) = E_r + jE_i$ [2,H,W].
- **Auxiliary targets:** S-parameters (complex-valued, per-port), port masks.
- **Conditioning:** wavelength (normalized scalar) + geometric design parameters.

4.2 4.2 Complex-Valued Physics UNet

- **Architecture:** U-Net with encoder–bottleneck–decoder + skip connections.
- **Complex layers:**
 - **ComplexConv2d:** $(W_r + jW_i)(x_r + jx_i) = (W_r x_r - W_i x_i) + j(W_r x_i + W_i x_r)$
 - **ComplexGroupNorm:** Joint normalization of real/imaginary parts.
 - **ModReLU:** $f(z) = \text{ReLU}(|z| + b) \cdot \frac{z}{|z|}$ — gates magnitude, preserves phase.
 - **ComplexAttention:** Q, K, V in complex domain.
- **Timestep conditioning:** Sinusoidal embedding + MLP, injected via AdaGN (adaptive group norm).
- **Physics feature embedding:** Helmholtz residual features computed inside the network as additional input channels.
- Discuss why complex-valued: phase is first-class citizen, avoids phase wrapping issues in real-valued models.

Architecture diagram showing the Complex UNet with: (1) input channels, (2) encoder/decoder blocks with complex layers, (3) skip connections, (4) timestep injection, (5) physics feature computation.

4.3 4.3 Flow Matching Formulation

- Conditional flow matching on the field channels $[\text{Re}(E_z), \text{Im}(E_z)]$ only.
- Geometry (ε , source mask) as conditioning — not noised.
- Interpolation: $x_t = (1 - (1 - \sigma_{\min})t)x_0 + tx_1$, with $\sigma_{\min} = 10^{-4}$.
- Time sampling: 50/50 mixture of uniform and Beta(2,2) $\rightarrow$ concentrates training near $t = 0.5$.
- ODE sampling at inference: Euler or adaptive solver over $t \in [0, 1]$.

4.4 4.4 Physics-Constrained Losses

4.4.1 4.4.1 Helmholtz Residual Loss

- Residual: $R = \nabla^2 E_z + k_0^2 \varepsilon E_z$ computed via 5-point finite-difference Laplacian.
- Applied to the model’s predicted $\hat{x}_1$ (endpoint estimate), not the noisy x_t .
- **Spatial masking:** Exclude PML regions, source regions, and **dielectric interface pixels** (where $|\nabla \varepsilon| > \tau$).
- **Interface masking rationale:** The collocated-grid Laplacian stencil produces $O(1)$ errors at sharp ε discontinuities (Si/SiO₂ boundaries). Masking these pixels drops the GT residual floor from ~ 0.17 to ~ 0.01 – 0.03 , giving a meaningful physics metric.
- **Normalization:** Divide by driving-term scale $\langle (k_0^2 \varepsilon E)^2 \rangle$ to make the loss dimensionless and grid-independent.
- **Time gating:** Only apply when $t \geq t_{\min}$ (e.g., 0.5) since low- t predictions are too noisy for meaningful physics evaluation.
- Optional t -power weighting: upweight high- t samples where field estimates are cleaner.

4.4.2 4.4.2 S-Parameter Loss

- S-parameters extracted via modal decomposition: project predicted field onto waveguide eigenmodes at each port.
- Loss on both magnitude and phase of S-parameters vs. FDTD ground truth.
- Discuss port mask conventions and multi-port handling (2-port, 3-port, 4-port devices).

4.4.3 4.4.3 Endpoint Loss

- Direct MSE between predicted and true fields at $t = 1$.
- Amplitude-weighted to focus on high-signal regions.

4.5 4.5 Multi-Loss Curriculum Training

- **Phase A** (epochs 0– N_A): FM loss only. Model learns basic field structure and velocity prediction.
- **Phase B** (epochs N_A – N_B): Add Helmholtz residual + endpoint loss. Physics constraints guide the model toward valid solutions.
- **Phase C** (epochs N_B –): Add S-parameter loss. Fine-tune port coupling accuracy.
- Each loss has independent warmup ramp to prevent gradient shock.
- Rationale: Introducing all losses simultaneously causes gradient conflict; curriculum stabilizes training.

Training curriculum timeline showing which losses are active at each phase, with warmup ramps. Could also show loss curves over training.

4.6 4.6 Data Augmentation

- D2 symmetry group (horizontal flip) applied on-the-fly during training.
- Exploits the vertical mirror symmetry of elongated photonic devices.
- Effectively doubles training data without additional FDTD computation.
- Note: Full D4 (rotations) not applicable due to rectangular grid and non-square domain.

5 5. Dataset

5.1 5.1 Device Types

- Table of all device types with: name, number of ports, key parameters, parameter ranges.
- Devices: S-bend, Y-branch, directional coupler (+ optionally straight waveguide, MMI, euler bend, circular bend, crossing).
- Silicon-on-insulator platform: $n_{\text{eff}} \approx 2.4$ (core), $n_{\text{clad}} = 1.444$, C-band.

5.2 5.2 FDTD Simulation Setup

- Meep FDTD solver, 2D TE polarization (E_z component).
- Resolution: pixels/ μm (discuss tradeoff with data size).
- PML absorbing boundaries.
- Eigenmode source excitation.
- Convergence criterion: field decay tolerance.
- Per-sample outputs: $\varepsilon(x, y)$, complex $E_z(x, y)$, source mask, port masks, S-parameters.

5.3 5.3 Dataset Statistics

- Total geometries, wavelengths per geometry, total simulations.
- Latin hypercube sampling of parameter space with half-pixel quantization.
- Train/val/test split by geometry ID (80/10/10) — no wavelength leakage.
- Data format: sharded NPZ files with JSON index.

- (1) Example device geometries (one per type) showing ε maps.
- (2) Corresponding $|E_z|$ fields.
- (3) Parameter distribution histograms.

6 6. Experiments

6.1 6.1 Training Setup

- Hardware: GPU type, number of GPUs, DDP setup.
- Optimizer: AdamW, learning rate schedule (warmup + cosine decay).
- Mixed precision (FP16 forward, FP32 losses).
- Hyperparameters: hidden size, channel multipliers, attention resolutions, batch size, epochs.
- Curriculum timing: Phase A/B/C epoch boundaries and loss weights.

6.2 6.2 Baselines

- **Real-valued UNet:** Same architecture but with standard real convolutions (ablation of complex layers).
- **No physics loss:** FM-only training (ablation of Helmholtz constraint).
- **No curriculum:** All losses from epoch 0 (ablation of phased training).
- **Direct regression:** UNet trained with MSE loss only (no flow matching).
- Optionally: compare against FNO or other neural operator baselines.

6.3 6.3 Evaluation Metrics

- **Field quality:** PSNR, SSIM, relative L_2 error (amplitude and phase separately).
- **Physics compliance:** Helmholtz residual ratio (with interface masking) — report GT floor and model value.
- **S-parameter accuracy:** Magnitude error (dB), phase error (degrees), per-port breakdown.
- **Generalization:** Per-device-type metrics, out-of-distribution performance.
- **Computational cost:** Inference time vs. FDTD, model parameter count, FLOPs.

6.4 6.4 Results

6.4.1 6.4.1 Field Prediction Quality

- Quantitative: Table of PSNR, SSIM, L_2 error per device type.
- Qualitative: Side-by-side comparison of GT vs. predicted fields (amplitude and phase).
- Show that complex UNet preserves phase structure better than real-valued baseline.

Grid of example predictions: rows = device types, columns = ε , $|E_z|$ GT, $|E_z|$ pred, phase GT, phase pred, error map.

6.4.2 6.4.2 Physics Compliance

- Helmholtz residual: model vs. GT floor, with and without interface masking.
- Show that physics loss reduces residual beyond what FM-only achieves.
- Spatial maps of residual — should be near-zero away from interfaces.

6.4.3 6.4.3 S-Parameter Accuracy

- Scatter plots: predicted vs. true $|S_{ij}|$ and $\angle S_{ij}$.
- Per-device breakdown (especially directional coupler which has strong wavelength dependence).
- Compare S-param accuracy with and without S-param loss.

6.4.4 6.4.4 Ablation Studies

- Complex vs. real-valued UNet.
- With vs. without Helmholtz residual loss.
- With vs. without curriculum (all losses from start vs. phased).
- Effect of interface masking on residual metric.
- Effect of time gating ($t_{\min}$) on physics loss effectiveness.
- Effect of t -power weighting on residual convergence.

6.4.5 6.4.5 Computational Efficiency

- Inference time: single sample, batched.
- Speedup factor vs. FDTD (expect 100–1000 $\times$).
- Training cost: GPU-hours, wall-clock time.

7 7. Discussion

- **Why complex-valued matters:** Phase preservation for coherent devices (interferometers, couplers). Real-valued models struggle with phase wrapping.
- **Physics loss as regularization:** The Helmholtz loss doesn't just improve metrics — it constrains the solution space to physically valid fields, improving generalization.
- **Interface masking insight:** Finite-difference Helmholtz residual has an inherent floor at dielectric interfaces due to stencil errors. Masking these pixels ($|\nabla \varepsilon| > \tau$) gives a meaningful metric. Discuss implications for any work using PDE residuals on discontinuous-coefficient problems.
- **Curriculum necessity:** Multi-task gradient conflicts between FM velocity loss and physics losses. Phased introduction prevents early-training instability.

- **Limitations:**
 - 2D only (TE polarization) — extension to 3D is future work.
 - Fixed grid resolution — does not adapt to feature size.
 - Single-frequency per sample (not broadband).
 - Trained on specific device families — generalization to arbitrary freeform geometries untested.
- **Broader impact:** Fast field prediction enables gradient-based inverse design, topology optimization, and automated PIC layout.

8 8. Related Work

- **Neural surrogates for EM simulation:** Review of CNN/UNet-based field predictors (cite relevant photonics ML papers).
- **Physics-informed learning for photonics:** PINNs applied to waveguides, resonators, etc.
- **Neural operators:** FNO, DeepONet applied to PDEs; any photonics applications.
- **Flow matching / diffusion for scientific computing:** Score-based models for PDE solutions, molecular dynamics, weather prediction.
- **Complex-valued neural networks:** History and recent applications in signal processing, MRI reconstruction, wave physics.
- **Inverse design of photonic devices:** Topology optimization, adjoint methods, and how surrogates could accelerate these.

9 9. Conclusion

- Summarize: complex-valued flow matching + Helmholtz constraint + curriculum training = fast, physics-compliant photonic field prediction.
- Restate key numbers: accuracy, speedup, number of device types.
- **Future work:**
 - 3D simulations (full vectorial fields).
 - Broadband prediction (multiple wavelengths in one forward pass).
 - Integration with inverse design / topology optimization loops.
 - Freeform geometry generalization (not just parameterized devices).
 - Larger models / more data scaling laws.

10 10. Appendix (Supplementary Material)

- **A. Full architecture details:** Layer counts, channel dimensions, attention configuration.
- **B. Device parameter ranges:** Complete table of all parameter ranges per device type.
- **C. Training hyperparameters:** Full table of all hyperparameters used.
- **D. Additional qualitative results:** More example predictions across device types and wavelengths.
- **E. Phase anchoring:** Details of the global phase alignment procedure used before computing phase metrics.
- **F. Modal S-parameter extraction:** Mathematical details of eigenmode projection for S-parameter computation.
- **G. Helmholtz residual derivation:** Units, interface masking threshold selection, GT floor analysis.